

Go Baby Go

First Year Project



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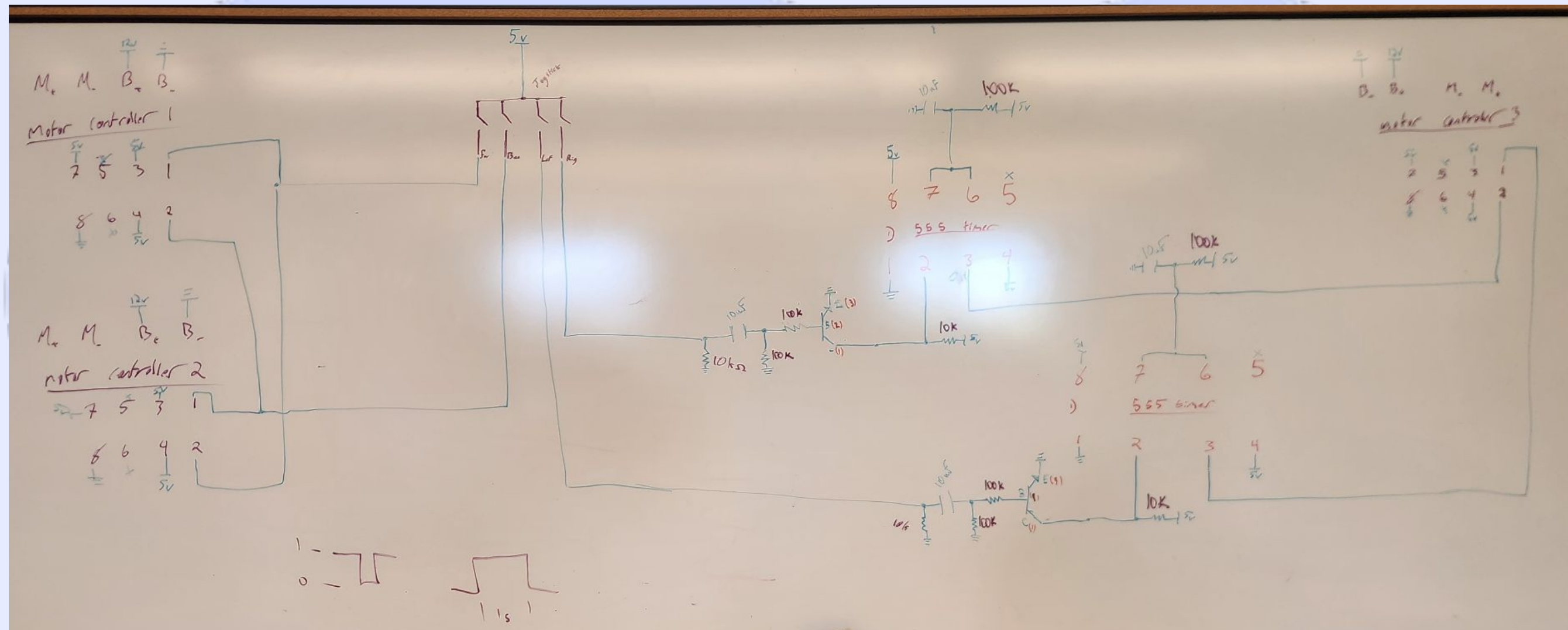
01. INTRODUCTION



What is *the Project*?

- This project was given to freshmen engineers as a first year project
 - Michael Riley (Civil Engineering), Francisco Cuen (Mechanical Engineering), Jaclyn Pellegrini (Mechanical Engineering), and Caleb Kasayka (Mechanical Engineering) make up Team Snow Munchers
 - Their task was to modify an electric car to use a joystick instead of foot pedal and steering wheel for Go Baby Go
- Other requirements included:
 - Directional Control
 - Consistent Steering & Speed Response
 - 12V Maximum Power
 - Dashboard Fit
 - Heat Limit



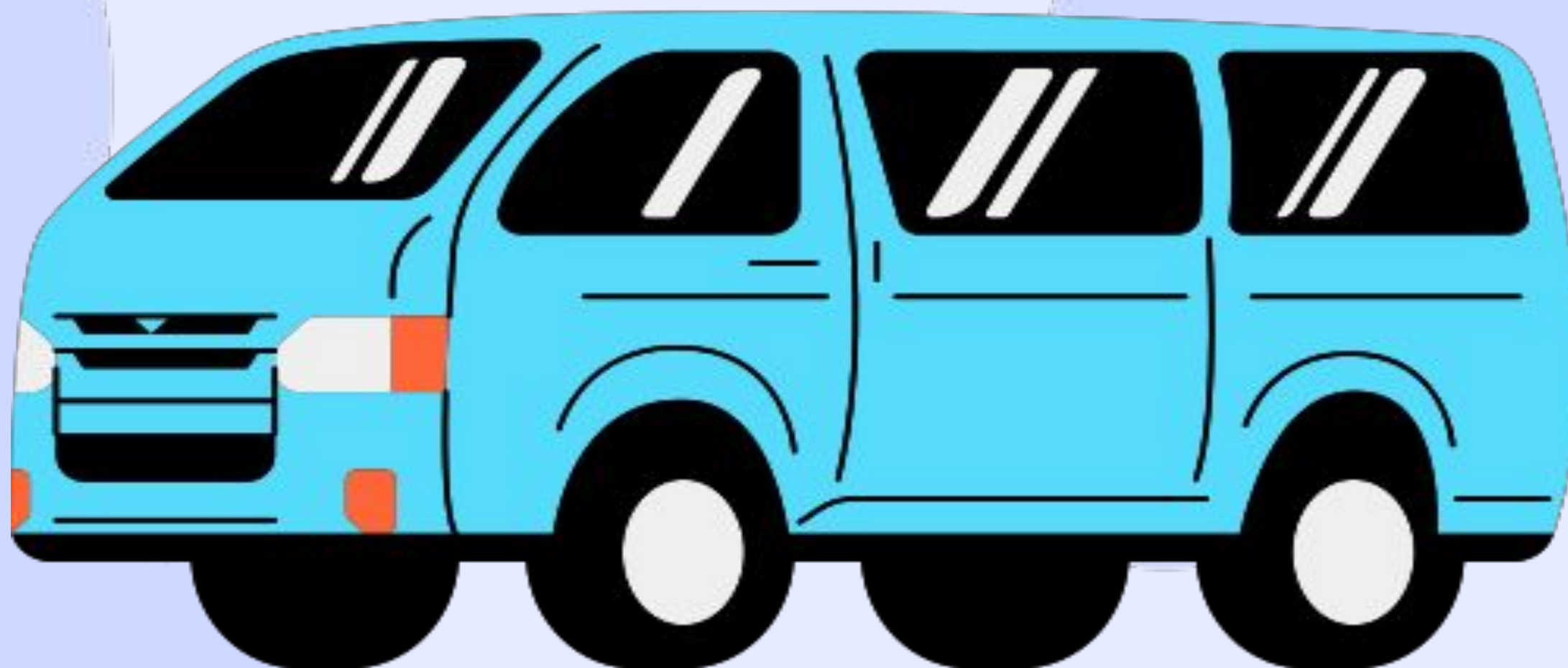


What is My Role in This?

Team Snow Munchers had designed a circuit for the modified electric car to run on. My role was to take the circuit, design a PCB for it, and to document the work on a GitHub repo.

02. My Role

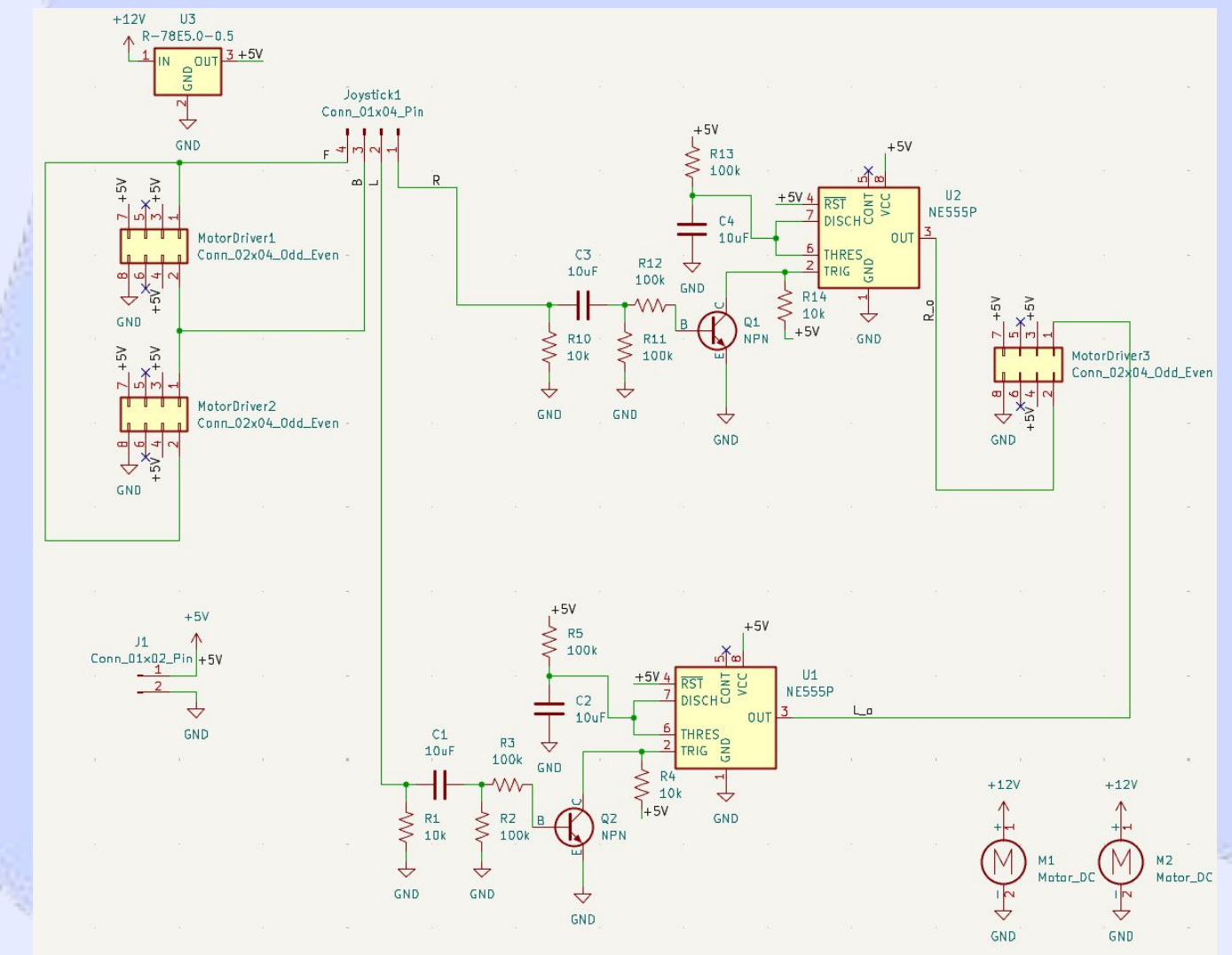
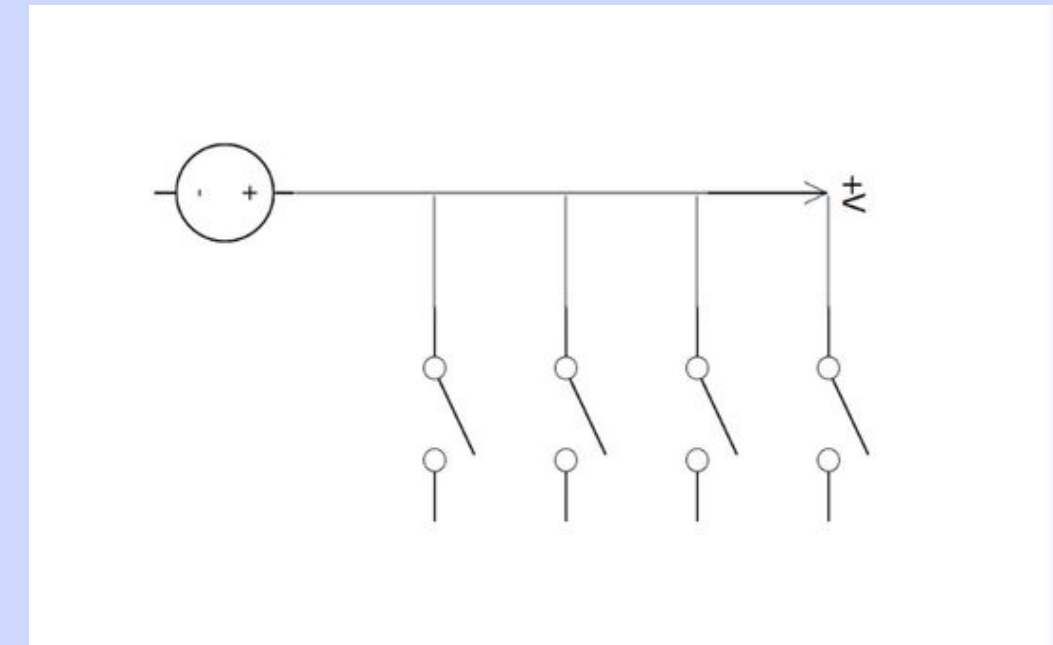
What I did with the project after the handoff



Circuit

Explanation

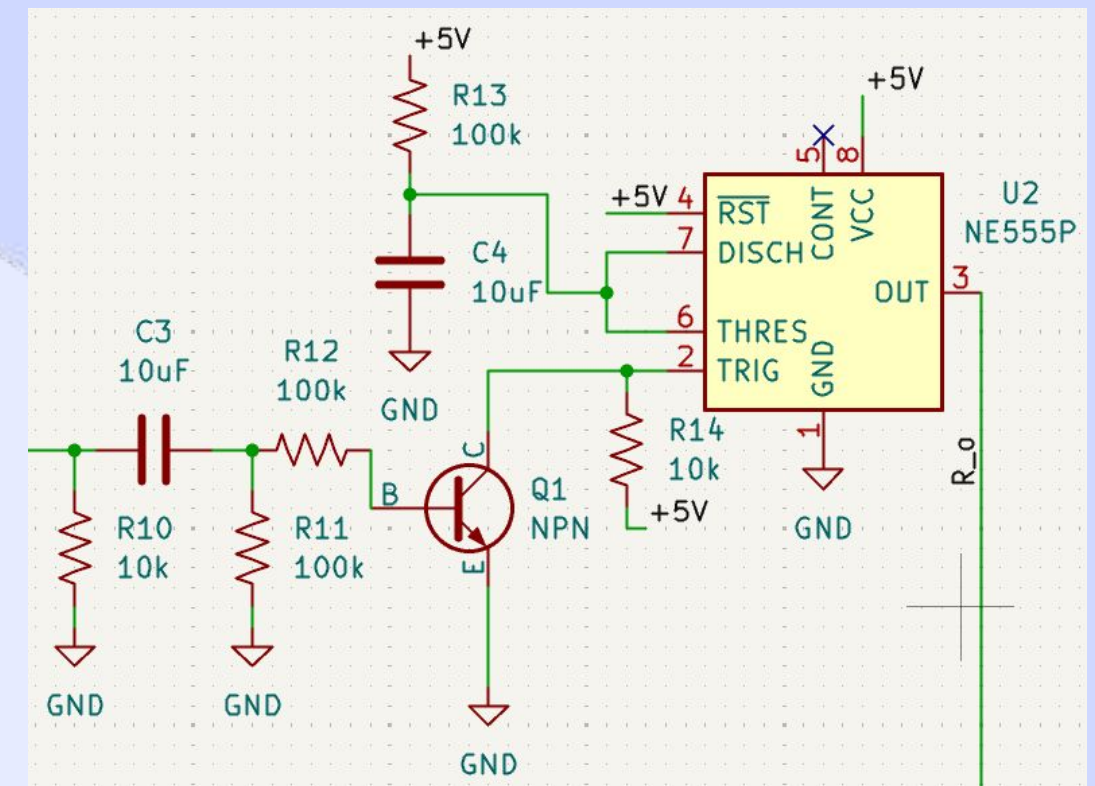
- Power comes from the 12V battery and is stepped down to 5V
 - Motors are still powered by 12V
- Everything is controlled by the joystick
 - Moving forwards, backwards, and turning left and right
- Each input from the joystick goes to a different path
 - 3 separate motor controllers work to move the motor forward, backwards, and to the sides
- While the forward and backwards controls go unrestricted, the turning could not
 - Needed a special fix



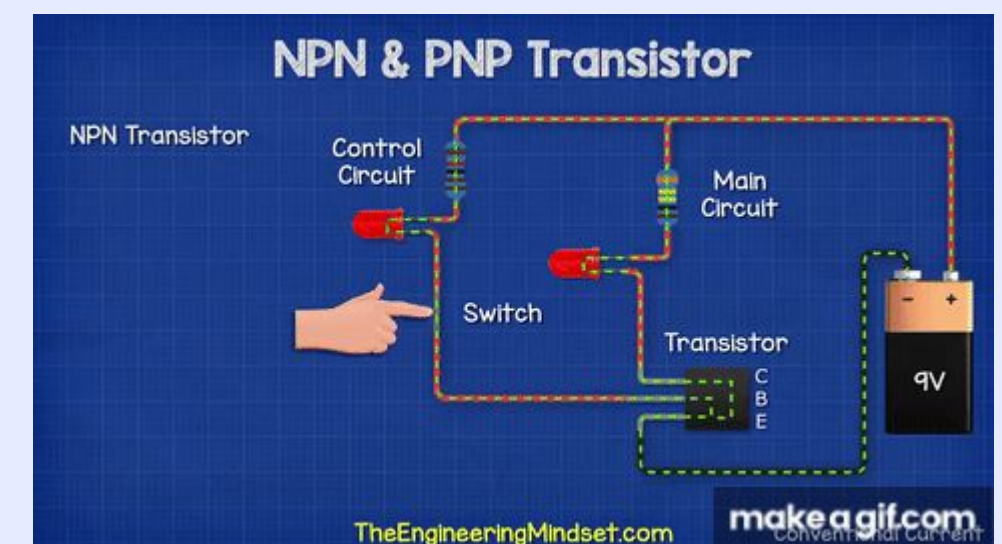
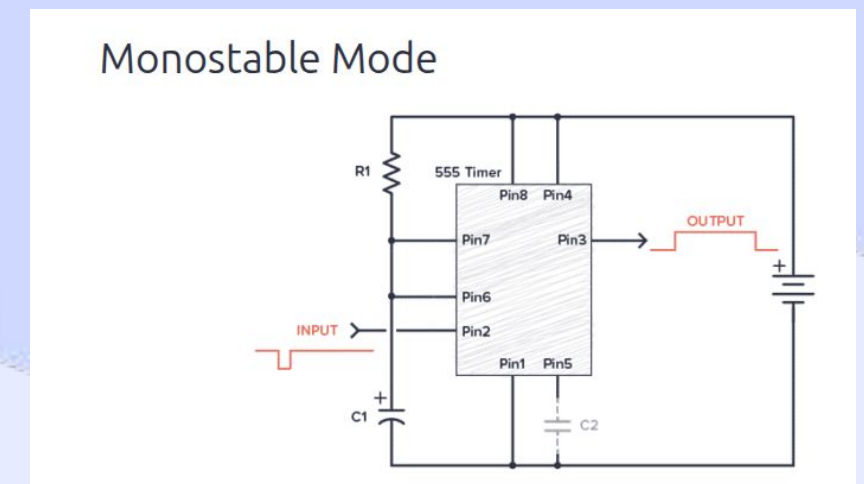
The

Timing Signal

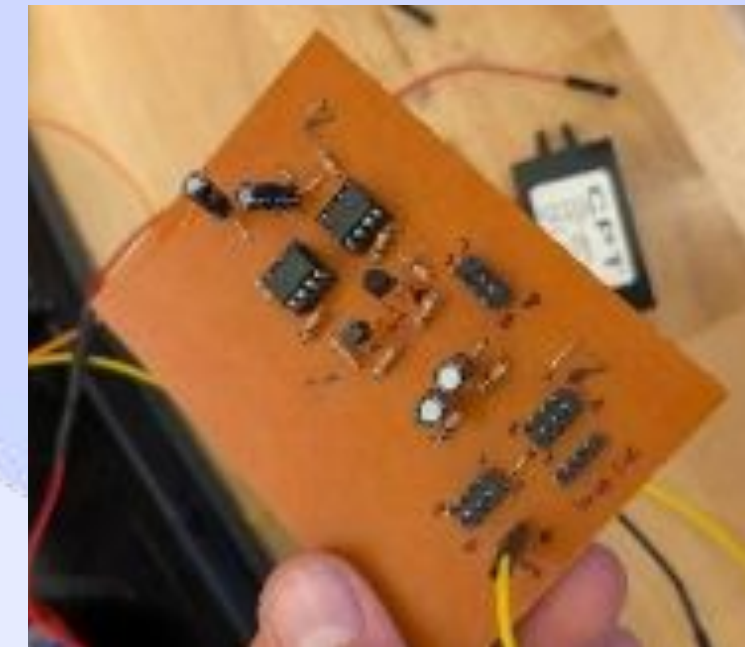
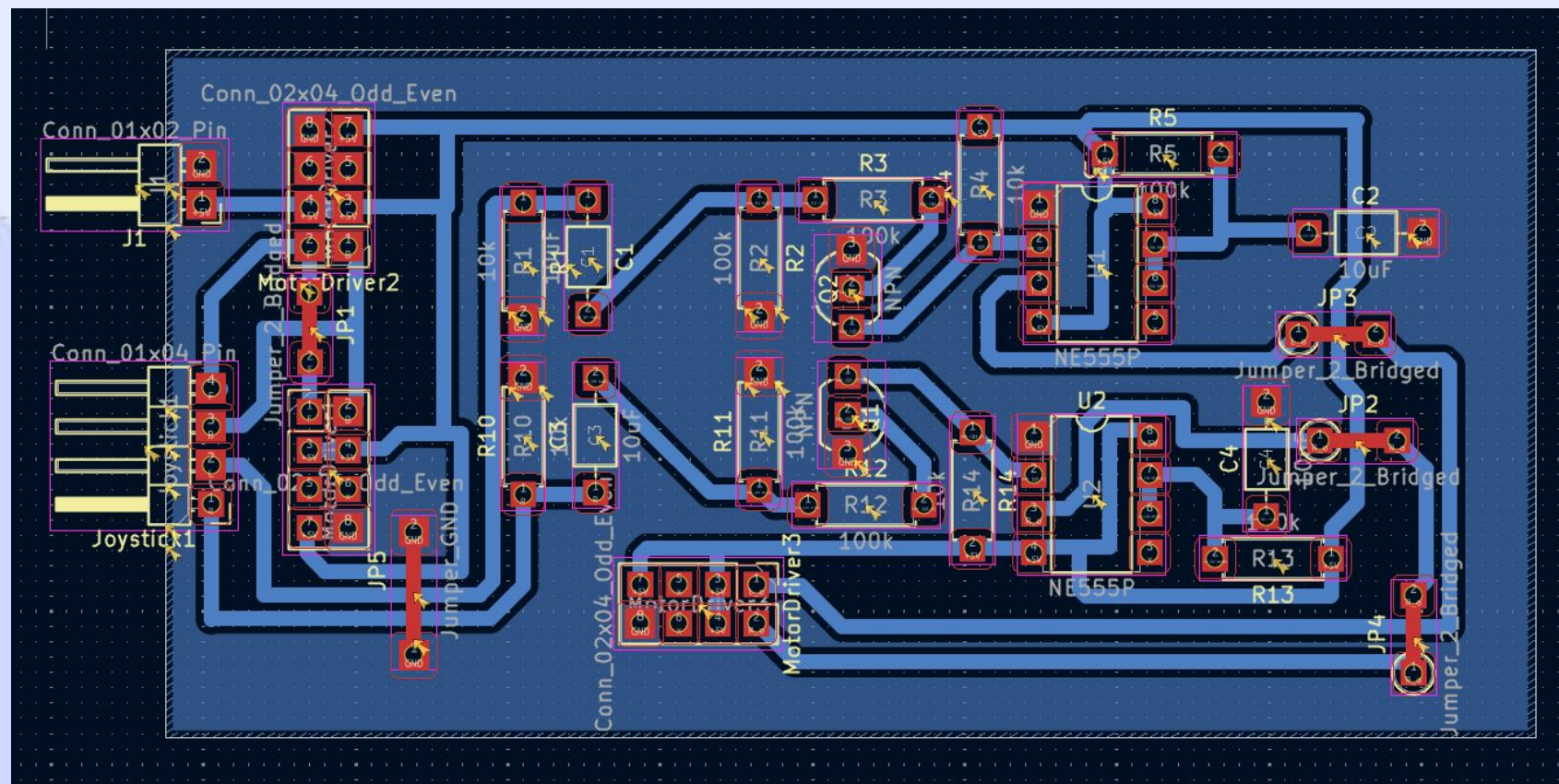
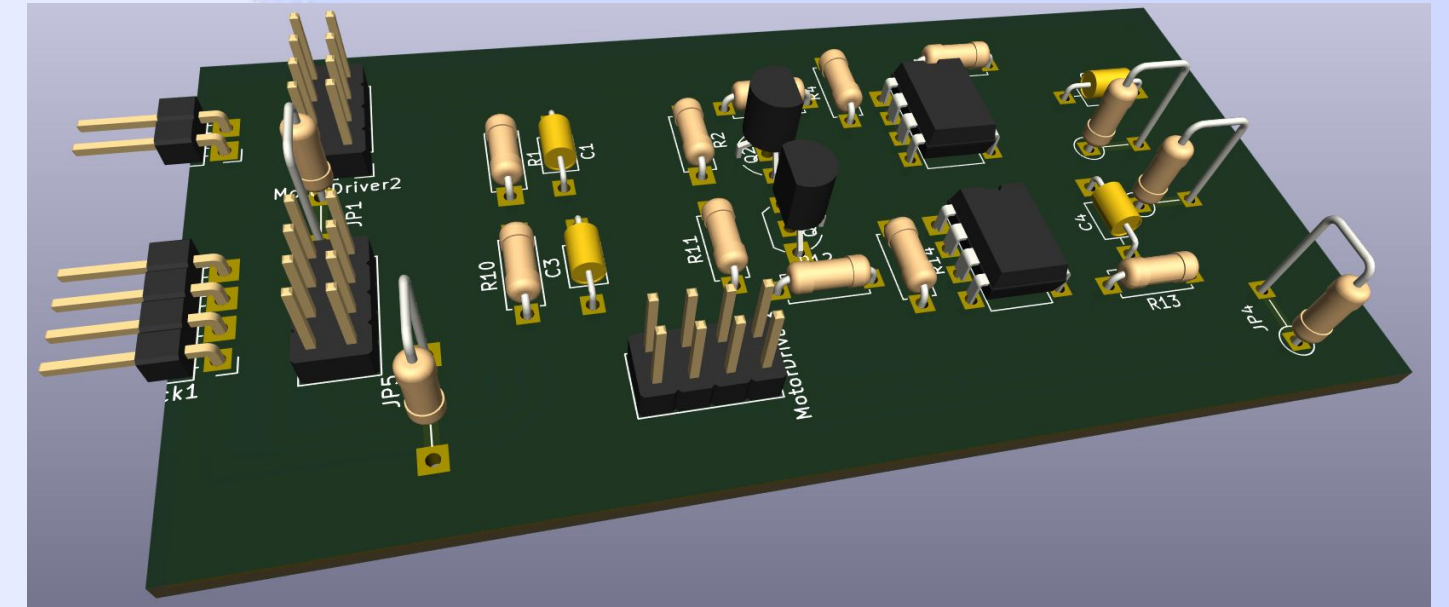
$$T = 1.1R_1C_1$$



- Having the left and right controls run without limit will cause the motors to burn out, so a different approach is needed:
 - A short pulse signal was generated to the motor driver that tells it to turn for ~1 sec
 - The resistor of 100kΩ resistor and a 10μF capacitor were used with the 555 timer to put it in monostable mode
 - The transistor acts as a switch once >0.7V is delivered to the Base (B) pin
 - Then the 555 timer signal is completed and a short 5V burst is sent to the motor controller



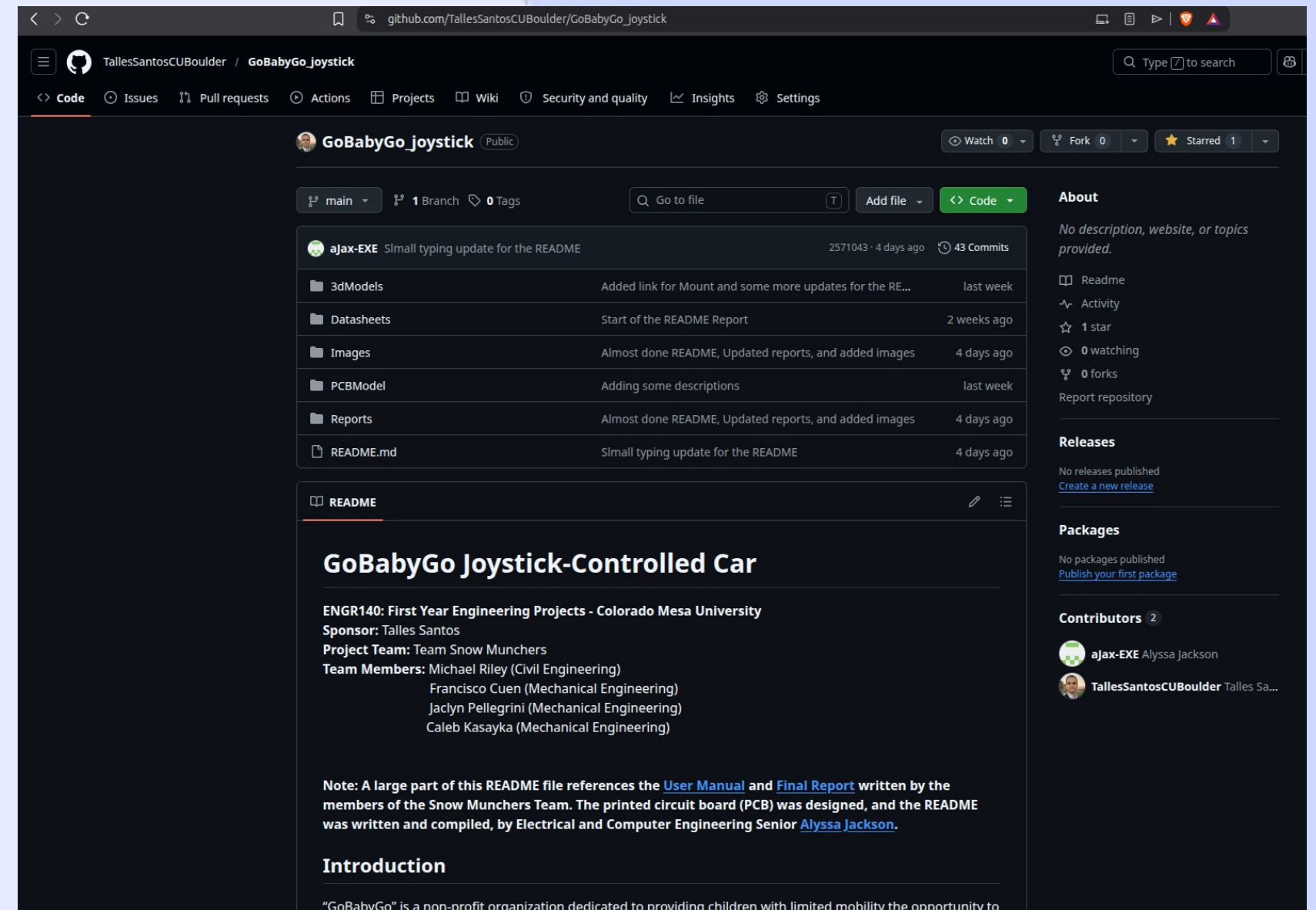
Circuit *Design*



Using the schematics drawn both on the board and modeled in KiCad, the PBC was designed, milled out, and soldered together

Writing the *GitHub*

- Another requirement was to write a GitHub
 - This project is meant to be open-sourced, so all documentation was uploaded to a GitHub repository to be used for free
 - PCB models, reports and instructions, images, datasheets, etc.
 - A README.md was written as a culmination of all of the work done for this project and explains everything to the readers step by step



03. Showing the *whip*



04.

Conclusions

- Overall this project wasn't for me
 - Yes it did combine what was taught in DSP, but it was designed to help out students who needed the help
- Their project won the First Year Engineering Project Judging Contest from Showcase
 - I consider that my win as well
- I'm proud of the students and excited to see where they go in the future



THANK *you!*

Any Questions?